

## 1.5 Practice - Formulas

Solve each of the following equations for the indicated variable.

1)  $ab = c$  for  $b$

3)  $\frac{f}{g}x = b$  for  $x$

5)  $3x = \frac{a}{b}$  for  $x$

7)  $E = mc^2$  for  $m$

9)  $V = \frac{4}{3}\pi r^3$  for  $\pi$

11)  $a + c = b$  for  $c$

13)  $c = \frac{4y}{m+n}$  for  $y$

15)  $V = \frac{\pi Dn}{12}$  for  $D$

17)  $P = n(p - c)$  for  $n$

19)  $T = \frac{D-d}{L}$  for  $D$

21)  $L = L_o(1 + at)$  for  $L_o$

23)  $2m + p = 4m + q$  for  $m$

25)  $\frac{k-m}{r} = q$  for  $k$

27)  $h = vt - 16t^2$  for  $v$

29)  $Q_1 = P(Q_2 - Q_1)$  for  $Q_2$

31)  $R = \frac{kA(T_1 + T_2)}{d}$  for  $T_1$

33)  $ax + b = c$  for  $a$

35)  $lwh = V$  for  $w$

37)  $\frac{1}{a} + b = \frac{c}{a}$  for  $a$

39)  $at - bw = s$  for  $t$

41)  $ax + bx = c$  for  $a$

43)  $x + 5y = 3$  for  $y$

45)  $3x + 2y = 7$  for  $y$

47)  $5a - 7b = 4$  for  $b$

49)  $4x - 5y = 8$  for  $y$

2)  $g = \frac{h}{i}$  for  $h$

4)  $p = \frac{3y}{q}$  for  $y$

6)  $\frac{ym}{b} = \frac{c}{d}$  for  $y$

8)  $DS = ds$  for  $D$

10)  $E = \frac{mv^2}{2}$  for  $m$

12)  $x - f = g$  for  $x$

14)  $\frac{rs}{a-3} = k$  for  $r$

16)  $F = k(R - L)$  for  $k$

18)  $S = L + 2B$  for  $L$

20)  $I = \frac{E_a - E_q}{R}$  for  $E_a$

22)  $ax + b = c$  for  $x$

24)  $q = 6(L - p)$  for  $L$

26)  $R = aT + b$  for  $T$

28)  $S = \pi rh + \pi r^2$  for  $h$

30)  $L = \pi(r_1 + r_2) + 2d$  for  $r_1$

32)  $P = \frac{V_1(V_2 - V_1)}{g}$  for  $V_2$

34)  $rt = d$  for  $r$

36)  $V = \frac{\pi r^2 h}{3}$  for  $h$

38)  $\frac{1}{a} + b = \frac{c}{a}$  for  $b$

40)  $at - bw = s$  for  $w$

42)  $x + 5y = 3$  for  $x$

44)  $3x + 2y = 7$  for  $x$

46)  $5a - 7b = 4$  for  $a$

48)  $4x - 5y = 8$  for  $x$

50)  $C = \frac{5}{9}(F - 32)$  for  $F$